IT-SIMPLERA SOLUTIONS

Network Administration Internship Program

WEEK 05 TASK REPORT

Enterprise Services Infrastructure Deployment
DHCP, DNS, HTTP/HTTPS, FTP, TFTP, NTP, Syslog, and SNMP

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1. Assignment Overview

This report documents the design, deployment, and verification of a centralized enterprise services infrastructure using Cisco Packet Tracer, completed as part of Week 5 of the Network Administration Internship Program at IT-Simplera Solutions.

The objective of this assignment was to implement and integrate multiple network services DHCP, DNS, HTTP/HTTPS, FTP, TFTP, NTP, Syslog, and SNMP across a multi-department enterprise environment, and to validate that all services operate together as a unified, reliable infrastructure.

2. Objectives

- Design and deploy a centralized enterprise services infrastructure using Cisco Packet Tracer to support multiple departments within an organization.
- Configure and integrate essential enterprise network services DHCP, DNS, HTTP/HTTPS, FTP, TFTP, NTP, Syslog, and SNMP to ensure reliable and efficient network operations.
- Implement secure access and service availability by applying appropriate configurations and enterprise networking best practices.
- Verify the functionality and interoperability of all configured services through comprehensive testing and connectivity validation.
- Troubleshoot and resolve service-related issues to maintain continuous network operation.

3. Tools Used

- Cisco Packet Tracer
- Cisco Router (2911)
- Cisco Catalyst 2960 Switches (x2)
- Server-PT (multi-service enterprise server)
- Desktop PCs (x8, distributed across departments)
- Command Prompt (ping, tracert, ftp, ipconfig)
- Web Browser (HTTP/HTTPS/DNS verification)

4. Network Topology

The network consists of one core router (Router0), two access-layer switches, one enterprise server (Server0), and eight client PCs distributed across two departments HR and Finance. Each department connects to the router through its own switch and subnet, while the enterprise server hosts all centralized services on a dedicated third subnet.

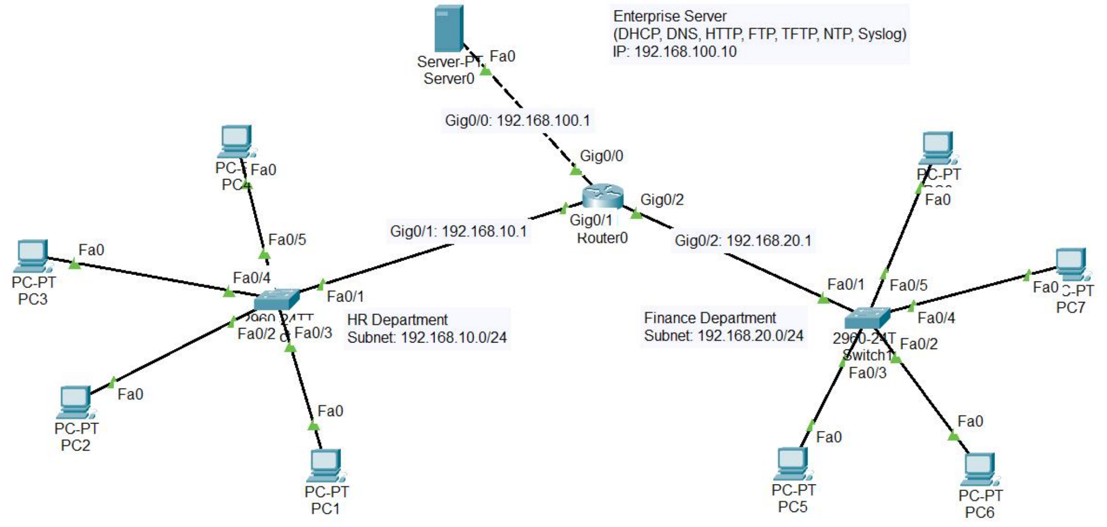


Figure 1: Enterprise Network Topology

4.1 IP Addressing Table

Device	Interface	IP Address	Notes
Router0	Gig0/0	192.168.100.1 /24	Server0
Router0	Gig0/1	192.168.10.1 /24	Switch0 (HR)
Router0	Gig0/2	192.168.20.1 /24	Switch1 (Finance)
Server0	Fa0	192.168.100.10 /24	Static, Gateway 192.168.100.1
PC1–PC4	Fa0	192.168.10.0/24 (DHCP)	HR Department — via Switch0
PC5–PC8	Fa0	192.168.20.0/24 (DHCP)	Finance Department — via Switch1

4.2 Router Interface Configuration

The following configuration was applied on Router0 to bring up all three interfaces and enable DHCP relay so that client requests originating on the HR and Finance subnets could reach the centralized DHCP server on the 192.168.100.0/24 network.

```
enable
configure terminal

interface GigabitEthernet0/0
 ip address 192.168.100.1 255.255.255.0
 no shutdown
 exit

interface GigabitEthernet0/1
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.100.10
```

```
 no shutdown
 exit

interface GigabitEthernet0/2
 ip address 192.168.20.1 255.255.255.0
 ip helper-address 192.168.100.10
 no shutdown
 exit

end
write memory
```

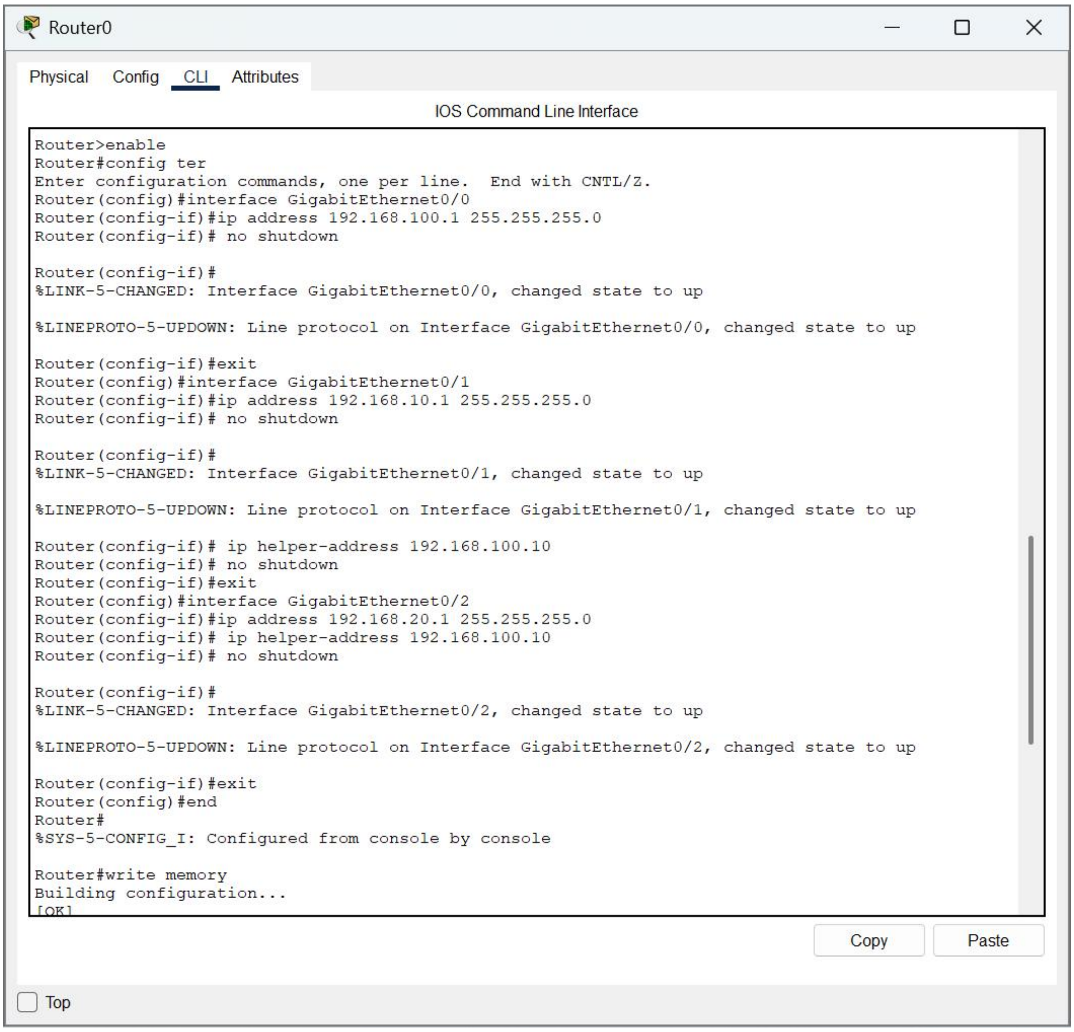


Figure 2: Router0 CLI — interface and DHCP helper-address configuration

5. Service Implementation and Concepts

This section explains each enterprise service that was implemented, the underlying networking concept, why it is required in an enterprise environment, the configuration applied, and the verification performed.

5.1 DHCP (Dynamic Host Configuration Protocol)

Concept

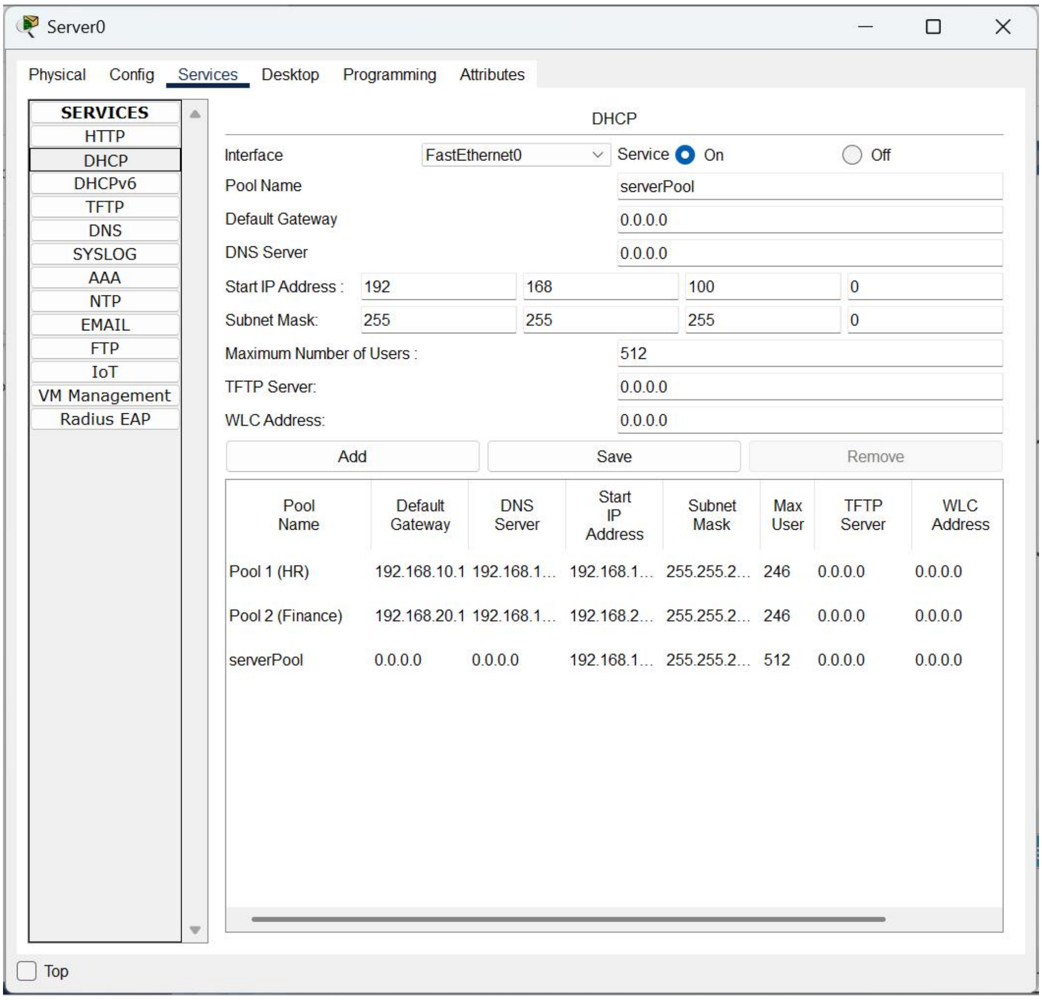
DHCP is a client–server protocol that automatically assigns IP addresses, subnet masks, default gateways, and DNS server information to hosts on a network, removing the need to configure each device manually.

Purpose in this Deployment

Two DHCP pools were created on Server0 — one for the HR subnet (192.168.10.0/24) and one for the Finance subnet (192.168.20.0/24). Because the DHCP server resides on a different subnet (192.168.100.0/24) than the clients, the `ip helper-address` command was applied on the router's HR and Finance interfaces to relay DHCP broadcast requests to the server.

Verification

- PCs on both departments were set to obtain an IP address automatically (DHCP mode).
- IP Configuration on client PCs confirmed successful assignment of IP address, subnet mask, and default gateway matching the correct department pool.



DHCP Pool Configuration on Server0

5.2 DNS (Domain Name System)

Concept

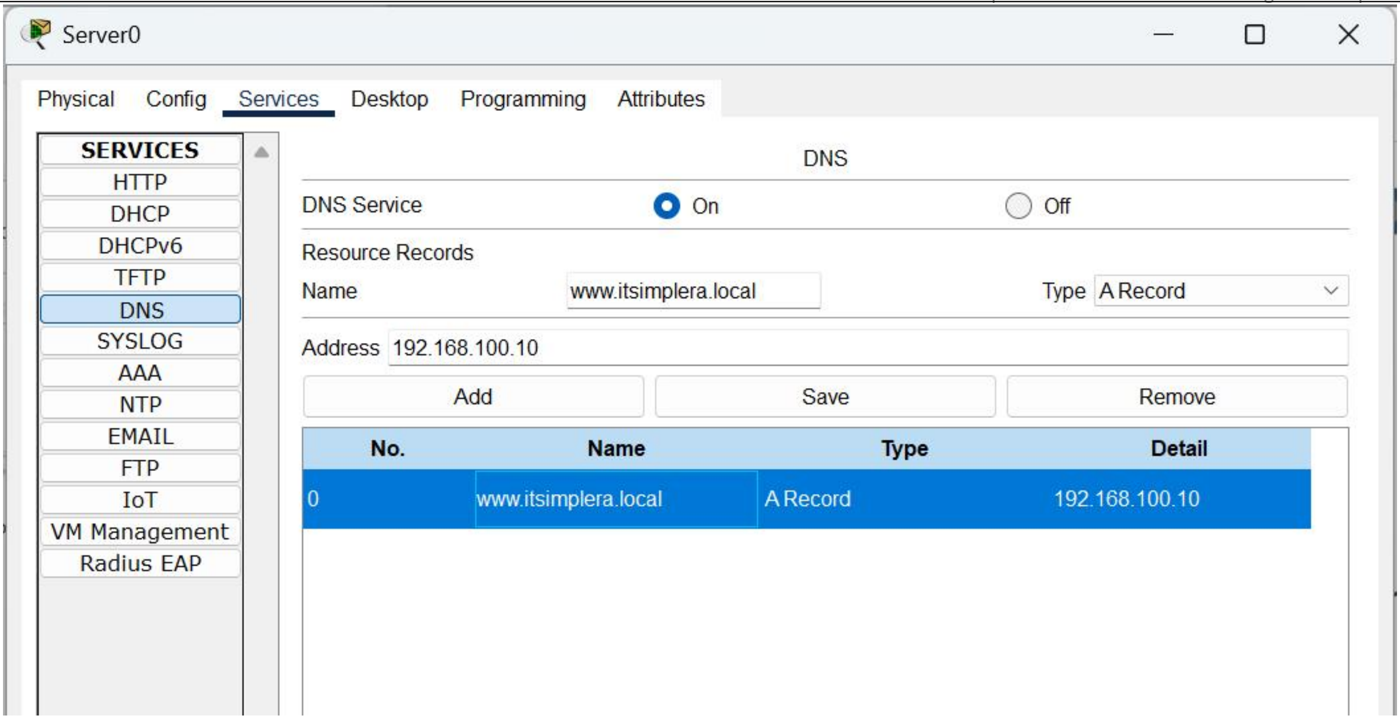
DNS translates human-readable hostnames (e.g. www.itsimplera.local) into IP addresses, allowing users to access network resources by name instead of memorizing numeric addresses.

Purpose in this Deployment

An A record was created on Server0 mapping the hostname www.itsimplera.local to the server's IP address 192.168.100.10, enabling name-based access to the enterprise web portal.

Verification

- From a client PC, the web browser was pointed to www.itsimplera.local and the enterprise portal loaded successfully, confirming correct name resolution.
- A ping to www.itsimplera.local also resolved to 192.168.100.10, confirming the DNS record was functioning.



DNS A Record Configuration on Server0

5.3 HTTP / HTTPS

Concept

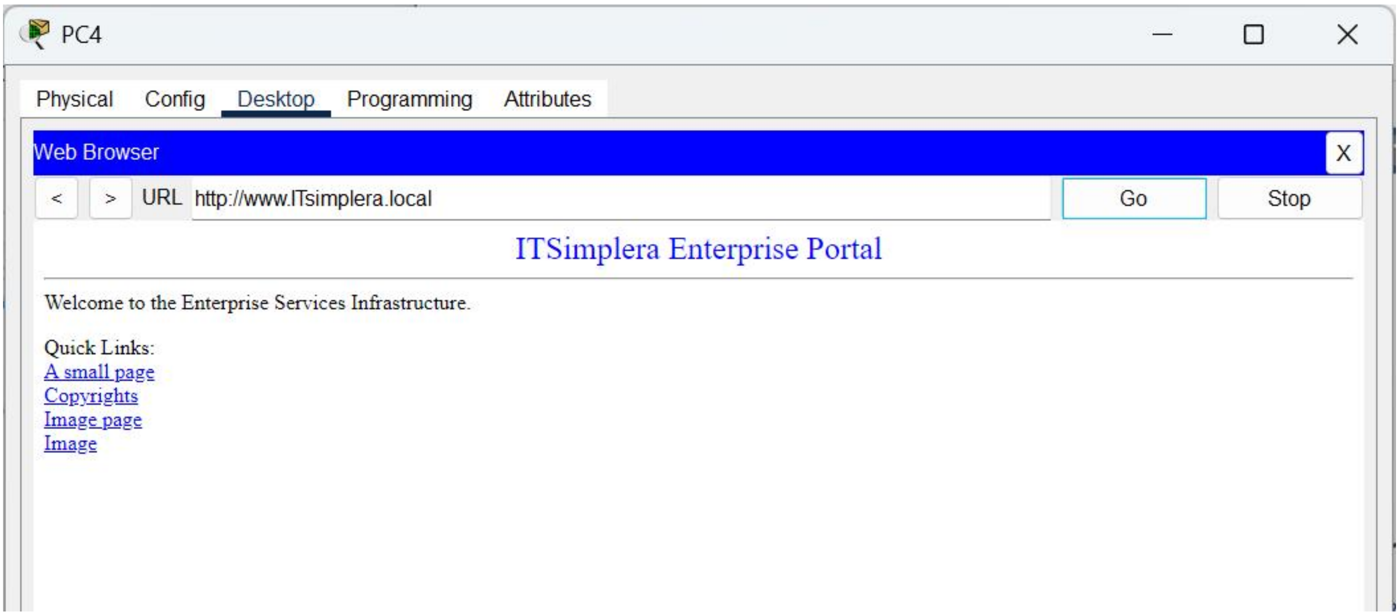
HTTP (and its secure version HTTPS) is the protocol used to serve and access web content. In an enterprise setting, an internal HTTP server commonly hosts a company portal, documentation, or internal tools accessible to employees.

Purpose in this Deployment

The HTTP/HTTPS service was enabled on Server0, and the default index.html page was edited to display an "ITSimplera Enterprise Portal" welcome page, confirming a working, customized enterprise web service.

Verification

- A client PC successfully accessed the portal via both the IP address and the DNS hostname, confirming end-to-end HTTP functionality.



HTTP Web Portal — Client Browser View

5.4 FTP (File Transfer Protocol)

Concept

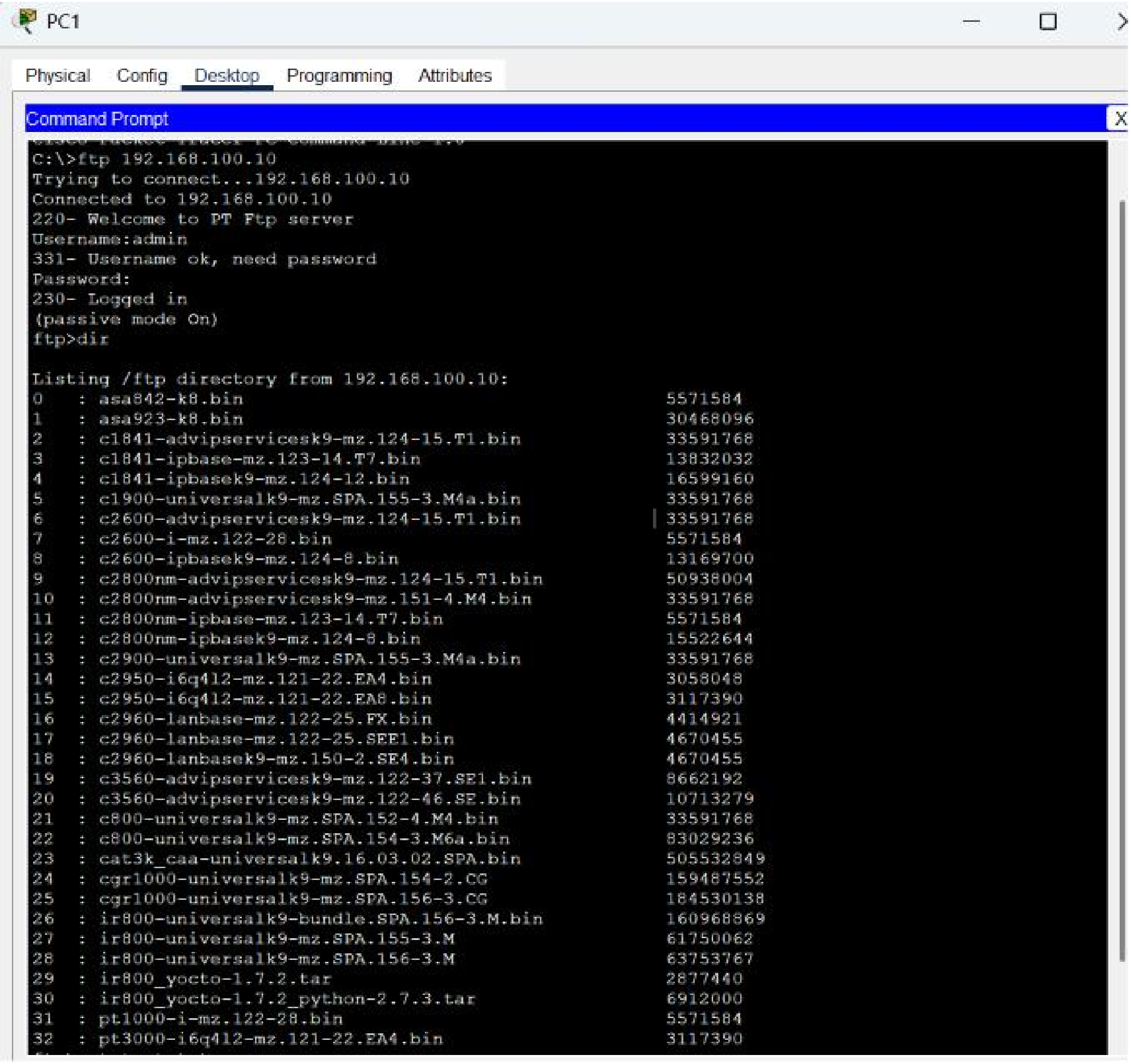
FTP is used to transfer files between a client and a server. It is commonly used in enterprise networks for centralized file sharing and distribution.

Purpose in this Deployment

An FTP user account was created on Server0 with read, write, delete, rename, and list permissions, allowing authenticated clients to upload and download files.

Verification

- From PC1's Command Prompt, the command `ftp 192.168.100.10` was used to connect to the server, log in with the configured credentials, and list the server directory using `dir`, confirming a successful FTP session.



FTP Login and Directory Listing from Client PC

5.5 TFTP (Trivial File Transfer Protocol)

Concept

TFTP is a lightweight, connectionless file transfer protocol most commonly used in enterprise networks to back up and restore device configuration files without requiring authentication.

Purpose in this Deployment

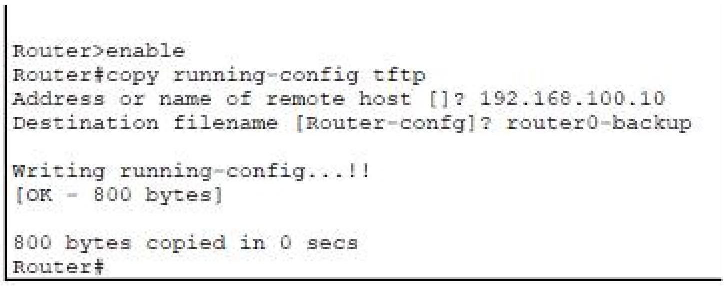
TFTP was enabled on Server0 and used to back up Router0's running configuration, demonstrating a standard enterprise configuration-backup workflow.

Configuration

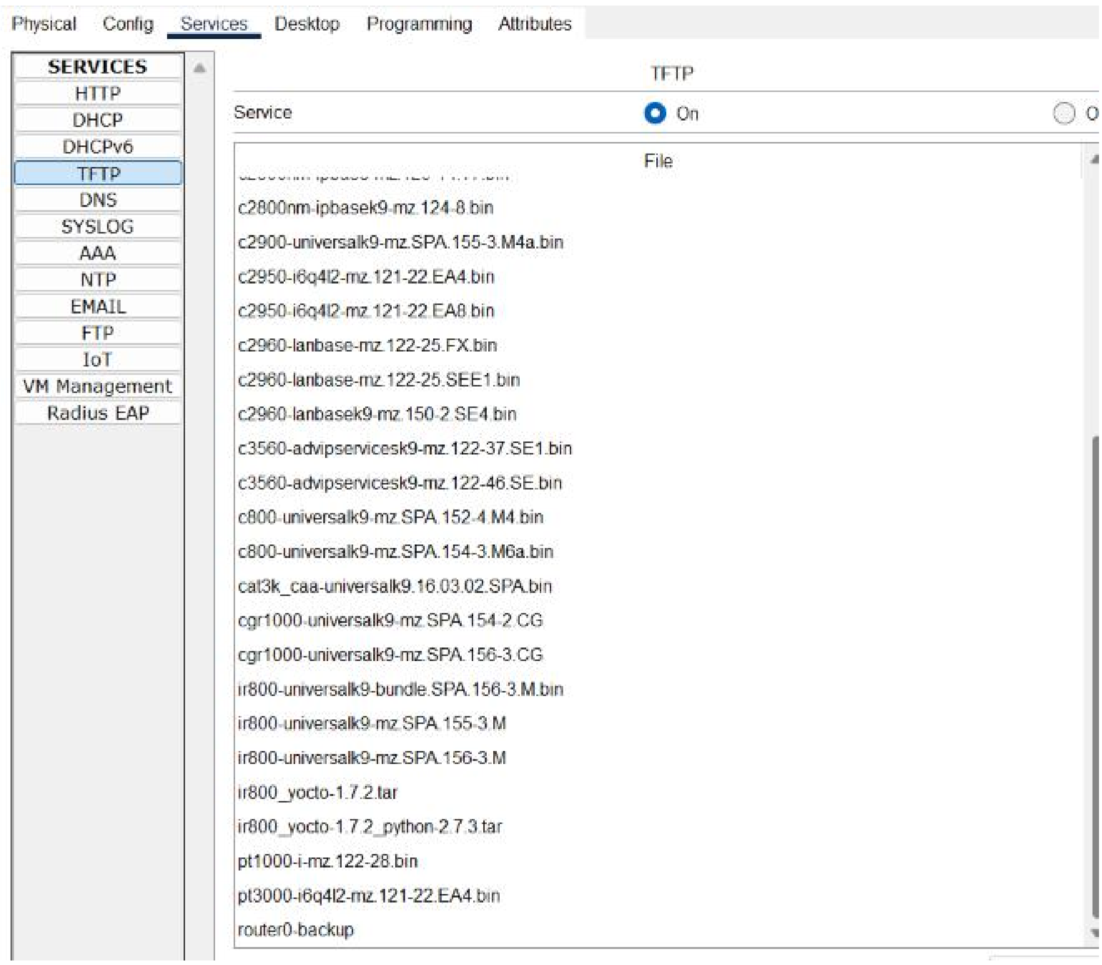
```
enable
copy running-config tftp
Address or name of remote host []? 192.168.100.10
Destination filename [Router-confg]? router0-backup
```

Verification

- The router confirmed the backup with an [OK] message and byte count.
- The backup file (router0-backup) was confirmed present in the TFTP file list on Server0.



Router Configuration Backup via TFTP (CLI)



Backup File Listed on Server0 (TFTP Service)

5.6 NTP (Network Time Protocol)

Concept

NTP synchronizes the clocks of network devices to a common time source, which is essential for accurate log timestamps, troubleshooting, and security auditing across an enterprise network.

Purpose in this Deployment

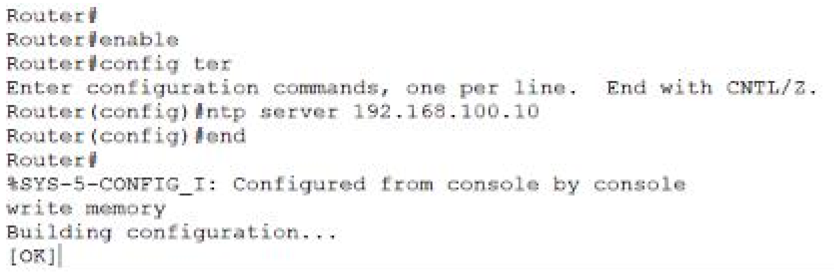
Server0 was configured as the NTP time source, and Router0 was configured as an NTP client, ensuring the router's clock remained synchronized with the centralized server.

Configuration

```
enable
configure terminal
ntp server 192.168.100.10
end
write memory
```

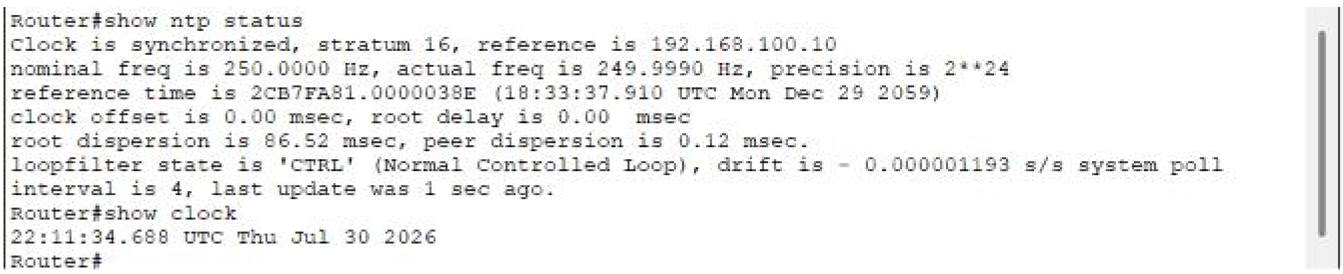
Verification

- `show ntp status` on Router0 confirmed the clock was synchronized to the NTP server at 192.168.100.10.
- `show clock` confirmed the router's system clock reflected the synchronized time.



```
Router#
Router#enable
Router#config ter
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ntp server 192.168.100.10
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
```

NTP Client Configuration on Router0



```
Router#show ntp status
Clock is synchronized, stratum 16, reference is 192.168.100.10
nominal freq is 250.0000 Hz, actual freq is 249.9990 Hz, precision is 2**24
reference time is 2CB7FA81.0000038E (18:33:37.910 UTC Mon Dec 29 2059)
clock offset is 0.00 msec, root delay is 0.00  msec
root dispersion is 86.52 msec, peer dispersion is 0.12 msec.
loopfilter state is 'CTRL' (Normal Controlled Loop), drift is - 0.000001193 s/s system poll
interval is 4, last update was 1 sec ago.
Router#show clock
22:11:34.688 UTC Thu Jul 30 2026
Router#
```

NTP Synchronization Verification (show ntp status)

5.7 Syslog

Concept

Syslog is a standard protocol used to forward log and event messages from network devices to a centralized logging server, allowing administrators to monitor and audit network activity from a single location.

Purpose in this Deployment

Router0 was configured to forward its log messages to Server0. An interface flap (shutdown/no shutdown) was performed on Gig0/1 to generate log events for verification.

Configuration

```
enable
configure terminal
logging 192.168.100.10
end
write memory
```

Verification

- The Syslog service on Server0 displayed the logging session start message ("Logging to host 192.168.100.10 port 514 started") confirming the router successfully connected.
- LINK-5-CHANGED and LINEPROTO-5-UPDOWN messages generated by the interface flap were received and displayed on the Syslog server, confirming centralized logging was fully functional.

```
Router#enable
Router#config ter
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#logging 192.168.100.10
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#write memory
Building configuration...
[OK]
```

Syslog Client Configuration on Router0

```
Router#config ter
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface gig0/1
Router(config-if)#shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
down

Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Interface Flap Test (Log-Generating Event)

Syslog

Service ◉ On ○ Off

	Time	HostName	Message
1	-	192.168.100.1	%SYS-5-CONFIG_I: Configured from console by console
2	-	192.168.100.1	%SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.100.10 port 514 started - CLI initiated
3	-	192.168.100.1	%SYS-5-CONFIG_I: Configured from console by console
4	-	192.168.100.1	%LINK-5-CHANGED: Interface GigabitEthernet0/1, ...
5	-	192.168.100.1	%LINEPROTO-5-UPDOWN: Line protocol on Interface ...
6	-	192.168.100.1	%LINK-5-CHANGED: Interface GigabitEthernet0/1, ...
7	-	192.168.100.1	%LINEPROTO-5-UPDOWN: Line protocol on Interface ...
8	-	192.168.100.1	%SYS-5-CONFIG_I: Configured from console by console

Syslog Messages Received on Server0

5.8 SNMP (Simple Network Management Protocol)

Concept

SNMP allows network devices to be monitored and managed by a centralized Network Management System (NMS), using community strings to control read-only or read-write access to device statistics.

Purpose in this Deployment

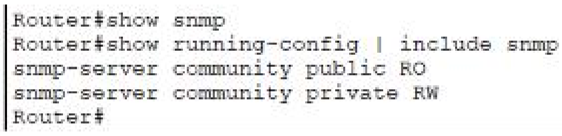
SNMP was enabled on Router0 with a read-only community string (public) and a read-write community string (private), establishing the router as an SNMP-manageable agent.

Configuration

```
enable
configure terminal
snmp-server community public RO
snmp-server community private RW
end
write memory
```

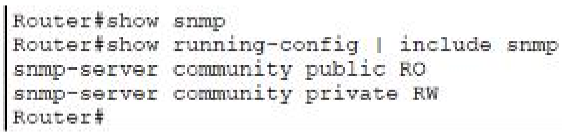
Verification

- `show running-config | include snmp` confirmed both community strings were successfully applied to the router configuration.
- Note: `show snmp` displays live polling statistics, which only populate when an active SNMP manager (NMS) is polling the device. Since Cisco Packet Tracer does not provide a built-in SNMP manager to generate polling traffic, configuration-level verification via running-config was used instead a valid and commonly used verification method when no external NMS is available.



```
Router#show snmp
Router#show running-config | include snmp
snmp-server community public RO
snmp-server community private RW
Router#
```

SNMP Configuration on Router0 (CLI)



```
Router#show snmp
Router#show running-config | include snmp
snmp-server community public RO
snmp-server community private RW
Router#
```

SNMP Configuration Verified via show running-config

6. Connectivity Testing

End-to-end connectivity was validated using ping and traceroute utilities from client PCs to confirm correct routing, DHCP addressing, and service reachability across the network.

6.1 Ping Tests

Test	Target	Result
Same-department (PC → PC)	192.168.10.11	Successful — 0% loss
Gateway reachability	192.168.10.1	Successful — 0% loss
Cross-department (HR → Finance)	192.168.20.11	Successful (minor initial timeout, then 0% loss)
Server reachability	192.168.100.10	Successful — 0% loss
DNS name resolution + ping	www.itsimplera.local	Resolved and successful — 0% loss

6.2 Traceroute Test

A traceroute from a client PC to the server (192.168.100.10) completed in two hops Router0's gateway interface, followed by Server0 — confirming correct routing through the core router.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping  192.168.10.11

Pinging 192.168.10.11 with 32 bytes of data:

Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128
Reply from 192.168.10.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.11

Pinging 192.168.20.11 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time<1ms TTL=127
Reply from 192.168.20.11: bytes=32 time=12ms TTL=127

Ping statistics for 192.168.20.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 4ms
```

```
C:\>ping 192.168.100.10

Pinging 192.168.100.10 with 32 bytes of data:

Reply from 192.168.100.10: bytes=32 time<1ms TTL=127
Reply from 192.168.100.10: bytes=32 time=8ms TTL=127
Reply from 192.168.100.10: bytes=32 time<1ms TTL=127
Reply from 192.168.100.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 8ms, Average = 2ms

C:\>ping www.itsimplera.local

Pinging 192.168.100.10 with 32 bytes of data:

Reply from 192.168.100.10: bytes=32 time=10ms TTL=127
Reply from 192.168.100.10: bytes=32 time=1ms TTL=127
Reply from 192.168.100.10: bytes=32 time<1ms TTL=127
Reply from 192.168.100.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.100.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>tracert 192.168.100.10

Tracing route to 192.168.100.10 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.10.1
  2   0 ms      0 ms      0 ms      192.168.100.10

Trace complete.
```

Figure 3: Ping and Traceroute verification from client PC

7. Troubleshooting Report

Issue Encountered	Diagnostic Steps / Commands Used	Resolution
snmp-server location / snmp-server contact commands rejected with "Invalid input" error.	Attempted with and without spaces, hyphens, and quotes.	Determined these optional metadata commands are not supported on this router's IOS image. Commands were skipped as they are non-essential to SNMP functionality; core community-string configuration was unaffected.
logging trap informational command rejected with "Invalid input" error.	Reviewed command syntax against IOS support in Packet Tracer.	Command omitted; default Syslog trap level was sufficient to generate and forward log messages successfully.
show snmp returned blank output despite successful configuration.	Verified configuration using show running-config | include snmp.	Confirmed this is expected behavior — show snmp only populates statistics when an active SNMP manager is polling the device, which Packet Tracer does not simulate. Verification was completed at the configuration level instead.
NTP client did not show "synchronized" immediately after configuration.	Waited and re-issued show ntp status after a short interval.	Clock synchronized successfully after allowing time for the NTP polling interval to complete.

8. Challenges Faced

- Understanding why DHCP requests required an ip helper-address on the router, since the DHCP server resided on a separate subnet from the client departments.
- Identifying that certain SNMP metadata commands (location, contact) were not supported on the router's IOS image in Packet Tracer, requiring verification through alternative means.
- Recognizing that show snmp requires active NMS polling traffic to display meaningful statistics a limitation of the simulated environment rather than a configuration fault.
- Allowing sufficient time for NTP synchronization to complete before verification.

9. Conclusion

This assignment successfully demonstrated the design and deployment of a centralized enterprise services infrastructure supporting multiple departments. DHCP, DNS, HTTP/HTTPS, FTP, TFTP, NTP, Syslog, and SNMP were configured, integrated, and verified to function together as a unified network. End-to-end connectivity testing confirmed reliable communication both within and across departments, and all encountered issues were diagnosed and resolved or appropriately documented. The completed topology reflects enterprise networking best practices and provides a solid foundation for further enhancements such as VLAN segmentation and dedicated network monitoring tools.